

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150NRL31HN0020ff gas separator7%.pvdb

Date of Analysis : Sep 16, 2026 11:31am

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 600.000 mm ID	
Vessel Design Length, Tangent to Tangent	1900.00 mm
Distance of Bottom Tangent above Grade	300.00 mm
Specified Datum Line Distance	0.00 mm
Internal Design Temperature	85 °C
Internal Design Pressure	2.500 MPa
External Design Temperature	85 °C
Maximum Allowable Working Pressure	3.9 MPa
Shop Test Pressure	5.069 MPa
Field Test Pressure	5.069 MPa
Required Minimum Design Metal Temperature	-40.0 °C
Warmest Computed Minimum Design Metal Temperature	-29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+N+IT
Head	SA-516 70	...	...	K02700	+N+IT
Nozzle	SA-350 LF1	1	...	K03009	+IT
Nozzle Flg	SA-350 LF1	1	...	K03009	+IT
Leg Baseplate	SA-36	...	...	K02600	
Leg Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Btm_head	2.506	0.00	3.900	1.5000	No	No
Shell	2.504	0.00	4.700	1.5000	N/A	No
Top_head	2.500	0.00	3.910	1.5000	No	No

Liquid Level: 875.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 0.676

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	50.0	50.0	12.0	10.0	6.9	...	1.00	1.00
Cylinder	1850.0	1800.0	12.0	12.0	7.0	...	1.00	1.00
Ellipse	1900.0	50.0	12.0	10.0	6.9	...	1.00	1.00

Local Stress Analysis Results:

Analysis	Max Stress	High Stress	Pass
----------	------------	-------------	------

Description	Type	Ratio	Location	Fail
B2_2"	WRC-107/537	0.902	n/a	Passed
A1_2"	WRC-297	0.819	Shell	Passed
B1_3"	WRC-107/537	0.900	n/a	Passed

Loads for Foundation/Support Design:

Total Wind Shear on top of all Legs	1632.	N
Total Earthquake Shear on top of all Legs	1385.	N
Total Wind Moment at top of all Legs	1142606.	N-mm
Total Earthquake Moment at top of all Legs	1310107.	N-mm
Wind Load Case Test Moment on Support	377060.	N-mm
Test Case Wind Shear on Support	538.	N
Max. Wind Shear on one Leg (top & bottom)	544.	N
Max. Earthq. Shear on one Leg (top & bottom)	462.	N
Max. Wind Moment at base of one Leg	753576.	N-mm
Max. Earthquake Moment at base of one Leg	639704.	N-mm
Max. Vertical Load (Wt. + Wind) on one Leg	6288.	N
Max. Vertical Load (Wt. + Eq.) on one Leg	6793.	N

For the Pressure Test Case:

Max. Wind Shear on one Leg (top & bottom)	179.	N
Max. Wind Moment at base of one Leg	248680.	N-mm
Max. Vertical Load (Wt. + Wind) on one Leg	5527.	N
Max. Vertical Load (Wt. + Eq.) on one Leg	4389.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	673.6	kgm
Shop Test - Fabricated + water (full)	1267.1	kgm
Shipping - Fab. + rem. intls.+ shipping app.	749.3	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	749.3	kgm
Empty - Fab. + intls. + details + wgths.	749.3	kgm
Operating - Empty + operating liquid (no ca)	906.9	kgm
Field Test - Empty weight + test medium (full)	1342.8	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
B2_2"	3.9	...	...	OK	4.92	OK	No Calc[*]
A1_2"	4.7	...	...	OK	4.92	OK	No Calc[*]
H1_6"	4.11	...	...	OK	8.66	OK	Passed
L1_2"	4.7	...	...	OK	6.30	OK	No Calc[*]
L3_2"	4.7	...	...	OK	6.30	OK	No Calc[*]
L2_2"	4.7	...	...	OK	6.30	OK	No Calc[*]
L4_2"	4.7	...	...	OK	6.30	OK	No Calc[*]
B1_3"	3.91	...	...	OK	6.30	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 3.9 Nozzle : B22"
 Minimum MAWP Shells/Flanges : 3.9 Element : Btmhead
 Minimum MAPnc Shells/Flanges : 4.58 Element : Tophead

Computed Vessel M.A.W.P. : 3.9 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
10	B2_2"	0.000	0.000	96.068
20	A1_2"	1050.000	180.000	96.068
20	H1_6"	200.000	0.000	310.800
20	L1_2"	800.000	0.000	107.600
20	L3_2"	800.000	45.000	107.600
20	L2_2"	400.000	90.000	107.600
20	L4_2"	400.000	45.000	107.600
30	B1_3"	0.000	0.000	144.862

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
B2_2"	2.000 in	160	LWN	2.375	8.738	...	...	261.50	600
A1_2"	2.000 in	160	LWN	2.375	8.738	...	...	214.75	600
L1_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	221.35	300
L3_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	221.35	300
L2_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	221.35	300
L4_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	221.35	300
B1_3"	3.000 in	160	WNF	3.500	11.125	...	...	212.74	600
H1_6"	6.000 in	Actual	WNF	6.945	12.000	...	...	232.19	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
B2_2"	SA-350 LF1	8.000	10.000	...	...	...
A1_2"	SA-350 LF1	8.000	10.000	...	...	...
L1_2"	SA-350 LF1	8.000	10.000	...	...	...
L3_2"	SA-350 LF1	8.000	10.000	...	...	...
L2_2"	SA-350 LF1	8.000	10.000	...	...	...
L4_2"	SA-350 LF1	8.000	10.000	...	...	...
B1_3"	SA-350 LF1	8.000	10.000	...	...	...
H1_6"	SA-350 LF1	8.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
B2_2"	...	0.0	250.00	0.00	Btm_head
A1_2"	1050.000	180.0	200.00	0.00	Shell
L1_2"	800.000	0.0	206.38	0.00	Shell
L3_2"	800.000	45.0	206.38	0.00	Shell
L2_2"	400.000	90.0	206.38	0.00	Shell
L4_2"	400.000	45.0	206.38	0.00	Shell
B1_3"	...	0.0	200.00	0.00	Top_head
H1_6"	200.000	0.0	200.00	0.00	Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 12.0mm THK X 600.0mm ID X 50.0mm	SA-516 70
1	CYLINDER: 12.0mm THK X 600.0mm ID X 1800.0mm	SA-516 70
2	INSULATION: 200mm X 70mm THK	
1	INSULATION: 1800mm X 70mm THK	
3	LEGS, SHAPE: L3X3X0.2500 1385.0mm	...
3	BASEPLATES: 154.0mm X 154.0mm X 14.0mm	SA-36
3	BASEPLATE BOLTS: 24.0mm	SA-36
1	CLASS 300 GR 1.4, 6.0" BLIND FLANGE(S)	SA-350 LF1
1	NAMEPLATE	...

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Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Btmhead	[10]	!	-40	-40		1.000	10.000	1.00	No
Btmhead	[7]	!	-40	-50		0.827	12.000	1.00	No
Shell	[8]	!	-40	-50		0.826	12.000	1.00	No
Tophead	[10]	!	-40	-40		0.999	10.000	1.00	No
Tophead	[7]	!	-40	-50		0.825	12.000	1.00	No
B22"	[1]	!	-29	-29	-29	0.902	10.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
A12"	[1]	!	-29	-29	-29	0.825	12.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
H16"	[1]	!	-29	-29	-29	0.826	12.000	1.00	No
Nozzle Flg	[5]	!	-29	-34					
L12"	[1]	!	-29	-29	-29	0.825	12.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
L32"	[1]	!	-29	-29	-29	0.825	12.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
L22"	[1]	!	-29	-29	-29	0.826	12.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
L4"2"	[1]	!	-29	-29	-29	0.826	12.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
B13"	[1]	!	-29	-29	-29	0.900	10.000	1.00	No
Nozzle Flg	[5]	!	-29	-68					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: -29 -29

Required Minimum Design Metal Temperature -40.0 °C

Warmest Computed Minimum Design Metal Temperature -29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

- Impact test temps were not entered in and not considered in the analysis.
- UCS-66(i) applies to impact tested materials not by specification and
- UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
- The Basic MDMT includes the (30F) PWHT credit if applicable.





Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	2.5	MPa
Design Internal Temperature	85.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-40.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW	
Load Case 17	NP+VO+OW	
Load Case 18	FS+BS+IP+OW	
Load Case 19	FS+BS+EP+OW	
Load Case 20	BL+IP+OW	
Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 mm
Distance Upwind of Crest	Lh	0 mm
Distance from Crest to the Vessel	x	0 mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm
Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	2.000	
Site Class	D	
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg
M.A.W.P + Static Head to Nozzle	Y	
Consider MAP New and Cold in Noz. Design	N	
Consider External Loads for Nozzle Des.	Y	
Use ASME VIII-1 Appendix 1-9	N	
Perform Blast Load Analysis	No	
Material Database Year	Current w/Addenda or Code Year	

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:



Element#	1/3
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Btmhead
Distance "FROM" to "TO"	50 mm
Inside Diameter	600 mm
Element Thickness	10 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.5 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	242 MPa
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	-150 mm
Height/Length of Liquid	200 mm
Liquid Density	0.0006749 kg/cm³
Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	-150 mm
Height/Length of Insulation	200 mm
Thickness of Insulation	70 mm
Density	0.0002 kg/cm³
Element From Node	10
Detail Type	Nozzle
Detail ID	B22"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	77.246 N
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1 +IT

Element#	2/3
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	1800 mm
Inside Diameter	600 mm
Element Thickness	12 mm



Internal Corrosion Allowance	1.5	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.5	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+N+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	675	mm
Liquid Density	0.0006749	kg/cm³

Element From Node	20	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1800	mm
Thickness of Insulation	70	mm
Density	0.0002	kg/cm³

Element From Node	20	
Detail Type	Nozzle	
Detail ID	A12"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	77.246	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1	+IT

Element From Node	20	
Detail Type	Nozzle	
Detail ID	H16"	
Dist. from "FROM" Node / Offset dist	150	mm
Nozzle Diameter	6	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	560.85	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1	+IT

Element From Node	20	
Detail Type	Nozzle	
Detail ID	L12"	
Dist. from "FROM" Node / Offset dist	750	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	101.01	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1	+IT

Element From Node	20	
Detail Type	Nozzle	
Detail ID	L32"	



Dist. from "FROM" Node / Offset dist	750	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	45.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	101.01	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	L22"	
Dist. from "FROM" Node / Offset dist	350	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	101.01	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	L4"2"	
Dist. from "FROM" Node / Offset dist	350	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	45.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	101.01	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	

Element From Node	20	
Detail Type	Leg	
Detail ID	LEGS	
Dist. from "FROM" Node / Offset dist	200	mm
Diameter at Leg Centerline	662.77	mm
Leg Orientation	1	
Number of Legs	3	
Section Identifier	L3X3X0.2500	
Length of Legs	1385	mm

Element From Node	20	
Detail Type	Weight	
Detail ID	InsClipwt	
Dist. from "FROM" Node / Offset dist	900	mm
Miscellaneous Weight	421.66	N
Offset from Element Centerline	0	mm

Element#	3/3	
Element From Node	30	
Element To Node	40	
Element Type	Elliptical	
Description	Tophead	
Distance "FROM" to "TO"	50	mm
Inside Diameter	600	mm
Element Thickness	10	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.5	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C



Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+N+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	30	
Detail Type	Insulation	
Detail ID	Ins: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	200	mm
Thickness of Insulation	70	mm
Density	0.0002	kg/cm ³
Element From Node	30	
Detail Type	Nozzle	
Detail ID	B13"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	3	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	140.53	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1	+IT

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Btm_head	...	...	50	...	50
Shell	...	...	1850	...	1800
Top_head	...	...	1900	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Btm_head		2.5058	12	1.5	600	138	1
Shell		2.5045	12	1.5	600	138	2
Top_head		2.5	12	1.5	600	138	3

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Btm_head		2.5	3.9	4.58	10	6.94847	1
Shell		2.5	4.7	5.39	12	7.03207	2
Top_head		2.5	3.91	4.58	10	6.93594	3

Minimum 3.9 4.26

Warning: A standard flange is governing the MAP(NC)!

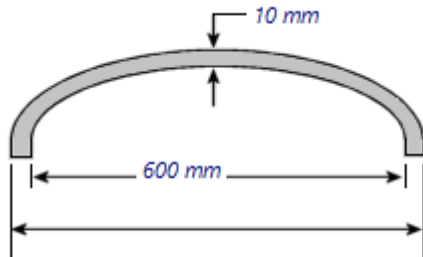
MAWP: 3.9 MPa, limited by: Btmhead.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Btmhead

Elliptical Head Design Information:					
Design Pressure	P	2.506 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	600.000 mm	Actual Stress		88.5 MPa
Spec. Min. Thk t _e , or s _a		10.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		520095.5 mm ²	SG of Contents		0.676
Element Volume		42.4 ltr	Weight of Contents		281.0 N
Empty Weight		496.5 N	Operating Weight		777.5 N
			Comp. All. Ope. Stress		137.0 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.750 mm

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.51 \cdot 603 \cdot 0.99) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 2.51)$$

$$= 5.4485 + 1.5000 = 6.9485 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.006 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 8.5) / (0.99 \cdot 603 + 0.2 \cdot 8.5)$$

$$= 3.91 - 0.0058 = 3.9 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 10) / (1 \cdot 600 + 0.2 \cdot 10)$$

$$= 4.58 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.51 \cdot (0.99 \cdot 603 + 0.2 \cdot 8.5)) / (2 \cdot 1 \cdot 8.5)$$



= 88.547 MPa

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)} \\ &= (2.51 \cdot 302) / (138 \cdot 1 - 0.6 \cdot 2.51) + 1.5 \\ &= 7.035 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.005 MPa

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (138 \cdot 1 \cdot 10.5) / (302 + 0.6 \cdot 10.5) \\ &= 4.71 - 0.0048 = 4.7 \text{ MPa} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6 \\ &= (2 + (603 / (2 \cdot 152)))^2 / 6 \\ &= 0.993416 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 8.333 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.500 mm

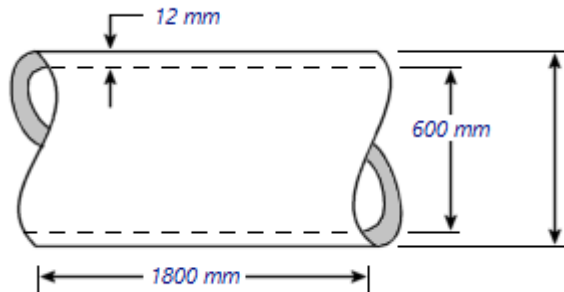
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.750 mm

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:					
Design Pressure	P	2.504 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		1800.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	600.000 mm	Actual Stress		73.4 MPa
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		3528636.8 mm²	SG of Contents		0.676
Element Volume		509.0 ltr	Weight of Contents		1264.6 N
Empty Weight		3156.1 N	Operating Weight		4420.7 N
			Comp. All. Ope. Stress		140.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.5 \cdot 302) / (138 \cdot 1 - 0.6 \cdot 2.5)$$

$$= 5.5321 + 1.5000 = 7.0321 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.004 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 10.5) / (302 + 0.6 \cdot 10.5)$$

$$= 4.71 - 0.0045 = 4.7 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 12) / (300 + 0.6 \cdot 12)$$

$$= 5.39 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.5 \cdot (302 + 0.6 \cdot 10.5)) / (1 \cdot 10.5)$$

$$= 73.417 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 1.961 %

Minimum Design Metal Temperature Results:



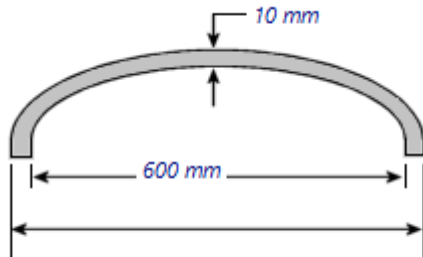
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Tophead

Elliptical Head Design Information:					
Design Pressure	P	2.500 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	600.000 mm	Actual Stress		88.3 MPa
Spec. Min. Thk t _e , or s _a		10.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		520095.5 mm ²	SG of Contents		0.000
Element Volume		42.4 ltr	Weight of Contents		0.0 N
Empty Weight		496.5 N	Operating Weight		496.5 N
			Comp. All. Ope. Stress		137.0 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.750 mm

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.5 \cdot 603 \cdot 0.99) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 2.5)$$

$$= 5.4359 + 1.5000 = 6.9359 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 8.5) / (0.99 \cdot 603 + 0.2 \cdot 8.5)$$

$$= 3.91 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 10) / (1 \cdot 600 + 0.2 \cdot 10)$$

$$= 4.58 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.5 \cdot (0.99 \cdot 603 + 0.2 \cdot 8.5)) / (2 \cdot 1 \cdot 8.5)$$

$$= 88.343 \text{ MPa}$$



Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (2.5 \cdot 302) / (138 \cdot 1 - 0.6 \cdot 2.5) + 1.5 \\ &= 7.022 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138 \cdot 1 \cdot 10.5) / (302 + 0.6 \cdot 10.5) \\ &= 4.71 \text{ MPa} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (603 / (2 \cdot 152))^2) / 6 \\ &= 0.993416 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 8.333 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.750 mm

Material treatment options: +N+IT

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5.069 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3.250 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	5.538 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	4.289 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	3.575 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3.575 MPa
App 27-4	= M.A.W.P.	3.899 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 3.9 • 1
 = 5.069 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Btm_head	153.2	234.0	0.655	5.09
Shell	130.3	234.0	0.557	5.09
Top_head	152.6	234.0	0.652	5.07

Theoretical Maximum Allowable test pressure is : 7.753 MPa

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
B2_2"	Nozzle	118.00	118.00	1.000
A1_2"	Nozzle	118.00	118.00	1.000
H1_6"	Nozzle	118.00	118.00	1.000
L1_2"	Nozzle	118.00	118.00	1.000
L3_2"	Nozzle	118.00	118.00	1.000
L2_2"	Nozzle	118.00	118.00	1.000
L4_2"	Nozzle	118.00	118.00	1.000
B1_3"	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Btm_head	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Top_head	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
B2_2"	18.05	184.50	0.098
A1_2"	18.01	184.50	0.098
H1_6"	35.36	184.50	0.192
L1_2"	10.75	184.50	0.058
L3_2"	10.75	184.50	0.058
L2_2"	10.76	184.50	0.058
L4_2"	10.76	184.50	0.058
B1_3"	21.13	184.50	0.115



Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5.069 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3.250 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	5.538 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	4.289 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	3.575 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3.575 MPa
App 27-4	= M.A.W.P.	3.899 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 3.9 • 1
 = 5.069 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Btm_head	179.9	234.0	0.769	5.09
Shell	149.2	234.0	0.638	5.09
Top_head	179.2	234.0	0.766	5.07

Theoretical Maximum Allowable test pressure is : 6.600 MPa

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
B2_2"	Nozzle	118.00	118.00	1.000
A1_2"	Nozzle	118.00	118.00	1.000
H1_6"	Nozzle	118.00	118.00	1.000
L1_2"	Nozzle	118.00	118.00	1.000
L3_2"	Nozzle	118.00	118.00	1.000
L2_2"	Nozzle	118.00	118.00	1.000
L4_2"	Nozzle	118.00	118.00	1.000
B1_3"	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Btm_head	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Top_head	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
B2_2"	18.05	184.50	0.098
A1_2"	18.01	184.50	0.098
H1_6"	35.36	184.50	0.192
L1_2"	10.75	184.50	0.058
L3_2"	10.75	184.50	0.058
L2_2"	10.76	184.50	0.058
L4_2"	10.76	184.50	0.058
B1_3"	21.13	184.50	0.115



Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	620	8.5	0.0019041	102.636
20	30	2000	624	10.5	0.00083663	79.6269
30	40	No Calc	620	8.5	0.0019041	102.636

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	10	No Calc	...	1.56
20	30	12	No Calc	...	1.79
30	40	10	No Calc	...	1.56

Minimum

1.56

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	2000	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	50.6305	42.4191	44.3017	42.9872	3.54413
20	30	321.857	509.029	282.315	514.132	22.53
30	40	50.6305	42.4191	44.3017	42.9872	3.54413
Total		423	593.87	370	600.11	29

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	28.6578	...	-75	...	Liquid: 10
10	Insl	10.3576	...	-50	...	Ins: 10
10	Nozl	8.42884	...	-320.006	...	B2_2"
20	Liqd	128.96	...	337.5	...	Liquid 20
20	Insl	55.007	...	900	...	Ins: 20
20	Nozl	8.42884	-400	1000	...	A1_2"
20	Nozl	61.1982	400	150	...	H1_6"
20	Nozl	11.0219	403.187	750	...	L1_2"
20	Nozl	11.0219	285.097	750	-285.097	L3_2"
20	Nozl	11.0219	...	350	-403.187	L2_2"
20	Nozl	11.0219	285.097	350	-285.097	L4_2"
20	Legs	40.3595	...	-492.5	...	LEGS
20	Wght	43	...	900	...	Ins_Clip_wt
30	Insl	10.3576	...	100	...	Ins: 30
30	Nozl	15.3341	...	295	...	B1_3"

Total Weight of Each Detail Type:

Liquid	157.6
Insulation	75.7
Nozzles	137.5
Legs	40.4
Weights	43.0
Sum of the Detail Weights	454.2 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	452.7	452.7	452.7	452.7	452.7	452.7
Nozzles	137.5	137.5	137.5	137.5	137.5	137.5
Legs	40.4	40.4	40.4	40.4	40.4	40.4
Wld Weights	43.0	43.0	43.0	43.0	43.0	43.0
Insulation	...	...	75.7	75.7	75.7	75.7
Ope. Liquid	...	...	...	...	...	157.6
Test Liquid	...	593.5	...	...	...	...
Totals	673.6	1267.1	749.3	749.3	749.3	906.9

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:



Fabricated Wt.	- Bare weight without removable internals	673.6 kgm
Shop Test Wt.	- Fabricated weight + water (full)	1267.1 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	749.3 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	749.3 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	749.3 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	906.9 kgm
Field Test Wt.	- Empty weight + test medium (full)	1342.8 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	188.8 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	306.6 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	209.3 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	520096	484482
20	30	3528636	3392920
30	40	520096	484482
Total		4568828.0	4361885.0 mm ²
Total		4.6	4.4 m ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	72.9611	101.619	115.354	...	72.9611
20	Legs	61.7898	76.1187	118.314	312312	61.7898
Legs	30	494.318	608.95	946.512	...	494.318
30	40	79.8663	79.8663	122.26	...	79.8663

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	...	...	...
20	Legs	-72.9611	-101.619	-115.354
Legs	30	574.185	688.816	1068.77
30	40	79.8663	79.8663	122.26

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	...	...	...
20	Legs	312312	312312	312312
Legs	30	...	...	...
30	40	...	...	...

Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
B2_2"	7.94	8.52	85	600	GR 1.4	6.337	5.577	7.943	5.577
A1_2"	7.94	8.52	85	600	GR 1.4	6.338	5.576	7.943	5.576
H1_6"	3.97	4.26	85	300	GR 1.4	...	...	...	...
L1_2"	3.97	4.26	85	300	GR 1.4	...	...	...	...
L3_2"	3.97	4.26	85	300	GR 1.4	...	...	...	...
L2_2"	3.97	4.26	85	300	GR 1.4	...	...	...	...
L4_2"	3.97	4.26	85	300	GR 1.4	...	...	...	...
B1_3"	7.94	8.52	85	600	GR 1.4	4.730	7.183	7.943	7.183
Min Rating	3.971	4.260, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
B2_2"	77.851	0.50	524985	3200	3.899	5.577	Pass
A1_2"	77.851	0.50	525024	3200	3.899	5.576	Pass
B1_3"	111.125	0.50	1158242	4200	3.899	7.183	Pass

Evaluating Flange on Nozzle [B22"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(524985) + 4 \cdot \text{abs}(3200) \cdot 77.851 \leq \pi \cdot 77.851^3 \cdot ((7.943 - 3.899) + .500 \cdot 7.943)$$

$$= 0.93966E+07 \leq 0.11884E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle B22" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(7.943, 0.500 \cdot 7.943 + 7.943 - 9396616.0 / (\pi \cdot 77.851^3))$$

$$= \min(7.943, 5.577)$$

$$= 5.577 \text{ MPa}$$

Evaluating Flange on Nozzle [A12"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(525024) + 4 \cdot \text{abs}(3200) \cdot 77.851 \leq \pi \cdot 77.851^3 \cdot ((7.943 - 3.899) + .500 \cdot 7.943)$$

$$= 0.93972E+07 \leq 0.11884E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A12" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(7.943, 0.500 \cdot 7.943 + 7.943 - 9397238.0 / (\pi \cdot 77.851^3))$$

$$= \min(7.943, 5.576)$$

$$= 5.576 \text{ MPa}$$

Evaluating Flange on Nozzle [B13"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1158242) + 4 \cdot \text{abs}(4200) \cdot 111.125 \leq \pi \cdot 111.125^3 \cdot ((7.943 - 3.899) + .500 \cdot 7.943)$$

$$= 0.20400E+08 \leq 0.34562E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle B13" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(7.943, 0.500 \cdot 7.943 + 7.943 - 20399544.0 / (\pi \cdot 111.125^3))$$

$$= \min(7.943, 7.183)$$



= 7.183 MPa

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 16.2728 Hz.

The Natural Frequency for the Vessel (Ope...) is 14.9532 Hz.

The Natural Frequency for the Vessel (Filled) is 12.0323 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.19 \cdot 0.98) / (1 + 1.7 \cdot 3.4 \cdot 0.19)))$
 $= \min(0.85, 0.91)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	14.953 Hz
Natural Frequency of Vessel (Empty)	16.273 Hz
Natural Frequency of Vessel (Test)	12.032 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.531
Structure Height to Diameter ratio	2.856
Height to top of Structure	2130.000 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$: $Zg = 589788$ mm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 159 + 300 = 459$ mm
 $= 1.51$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (1.51 ft.) < 15 ft.
 $= 2.41(15 / Zg)^{2/\alpha}$
 $= 2.41(15 / 589788)^{2/11.5}$
 $= 1.04$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.04 \cdot 1 \cdot 1 \cdot 1 \cdot 112^2)$$

$$= 33.1 \text{ psf [1.59] kPa}$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.1 \cdot 0.85 \cdot 0.53 \cdot 2.26$$

$$= 33.9 \text{ lbs. [151] N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Btmhead	459.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell	1400.0	0.000	0.000	0.000	1.035	1.000	1.587
Tophead	2421.0	0.000	0.000	0.000	1.035	1.000	1.587

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(104)$
 L - Total height of structure mm
 Lc - Total length of conical section(s) of vessel, mm
 tb - Uncorroded plate thickness at bottom of vessel, mm
 V30 - Design Wind Speed provided by user, km/hr
 Vc - Critical wind velocity, km/hr
 Vw - Maximum wind speed at top of structure, km/hr
 W - Total corroded weight of structure, N
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
 Z - Maximum amplitude of vibration at top of vessel, mm
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, <= 0.31416E-05 (High); 0.31416E-05 < 0.39270E-05 (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0/1900 = 0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[2.51/6.23^2] \cdot 10^4 = 645$ [Geometry Violation]

Compute the vibration possibility. If Vp > 0.39270E-05 no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 8345 / (1900 \cdot 717^2)$$

$$= 0.85410E-05$$

Since Vp is > 0.39270E-05 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
----	------------------------------	-----------------	-----------------	------------

InsClipwt 0.00 1250.00 1.59 0.00

The Natural Frequency for the Vessel (Ope...) is 14.9532 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	458.966	912	210345	1.58714	165.726
20	30	1400	916.8	1650240	1.58714	1300.18
30	40	2421.03	912	210345	1.58714	165.726

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio	z/h
Amplification Factor	Ap
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR]
Component strength factor	[Rpo]
Overstrength factor	[Omega0]
Response Acceleration Factor	[SMS]
Response Acceleration Factor	[SM1]
Building Importance Factor	[Ie]
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Seismic Analysis Results:

$$\begin{aligned} \text{SDS} &= 2/3 * \text{Sms} = 2/3 * 0.33 = 0.22 \\ \text{SD1} &= 2/3 * \text{Sm1} = 2/3 * 0.14 = 0.093 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= \text{Ct} \cdot \text{hn}^x$ where $\text{Ct} = 0.020$, $x = 0.75$ and $\text{hn} = \text{Structural Height (ft.)}$
 $= 0.020(10.40.75)$
 $= 0.116 \text{ seconds}$

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/15)$
 $= 0.067$

Check the Value of T which is the smaller of $\text{Cu} \cdot \text{Ta}$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.12, 0.067) \text{ per 12.8.2}$
 $= 0.067$

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 $= \text{Sds} / (R/I)$
 $= 0.22 / (2/1.5)$
 $= 0.163$

Maximum value of Cs per equation 12.8-3 [Cs]:
 $= \text{Sd1} / (T(R/Ie))$
 $= 0.093 / (0.067 (2/1.5))$
 $= 1.039$

Minimum value of Cs per equation 15.4-1 [Cs]:
 $= \max(0.044 \cdot \text{SDS} \cdot \text{Ie}, 0.030)$
 $= \max(0.044 \cdot 0.22 \cdot 1.5, 0.030)$
 $= 0.030$

Total Base Shear [V]:
 $= \text{Cs} \cdot \text{W}$ (Equation 12.8-1):
 $= 0.16 \cdot 8497$
 $= 1385.079 \text{ N}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot \text{Sds} \cdot \text{W}$
 $= 0.2 \cdot 0.22 \cdot 8497$



= 369.4 N

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related
 to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 14.9532 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	25	996.475	4.3403	3.01272	-43.3134
20	Legs	250	746.42	32.5115	25.5143	-75.7578
Legs	30	1050	5971.36	1092.39	857.281	293.597
30	40	1875	783.169	255.841	247.339	34.0417



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	321.034	...	...	...	...
20	Legs	100	165.726	4.3403	20066.5	525.536
Legs	30	800	1631.63	1385.08	1142606	1310107
30	40	1721.03	165.726	255.841	20066.5	30978

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	...	4.58818	4.58818	0.004968
20	Legs	165.726	4.58821	4.58832	0.0049693
Legs	30	1631.63	4.59119	4.59573	0.0049739
30	40	165.726	4.59588	4.59602	0.0049739

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	244.81	1530.06
20	30	246.099	1538.12
30	40	244.81	1530.06

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 9.500 Actual deflection : 4.596 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	19163.7	16329.2	2876101	2462577
20	30	23071.9	20237.3	3463439	3052566
30	40	19163.7	16329.2	2876101	2462577

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Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	165.6	280.8	-136.988	-136.988
20	Legs	165.597	280.795	-140.649	-140.649
Legs	30	165.597	280.795	-140.649	-140.649
30	40	165.597	280.795	-136.988	-136.988

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	43.8382	...	88.8914
20	30	35.3929	...	71.7666
30	40	43.8382	...	88.8914

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	...	...	...	...	...
20	Legs	0.035356	0.049244	0.0559	0.10228	0.10228
Legs	30	-0.27824	-0.2713	-0.27824	...	...
30	40	-0.047965	-0.047965	-0.047965	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	...	...	...	...	...
20	Legs	0.10228	0.0065715	0.00017211	0.0021686	...
Legs	30	...	0.37419	0.42904	0.12348	...
30	40	...	0.008146	0.012575	0.0026882	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	...
20	Legs	...	...	...	0.00011946
Legs	30	...	...	...	0.363
30	40	...	...	...	0.012158

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	...	-0.0026527
20	Legs	...	-0.0037438
Legs	30	...	0.014509
30	40	...	0.0020849

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	...	...	...	...
20	Legs	...	...	...	...
Legs	30	...	...	...	...
30	40	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	Legs	...
Legs	30	...
30	40	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	136.99	0.0000	-0.0000
20	0.14	165.60	-0.07	140.65	0.0009	0.0005
20	0.10	165.60	-0.65	140.65	0.0006	0.0046
30		165.60	-0.06	136.99		0.0004

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-0.00	136.99		0.0000
20	0.13	165.60	-0.07	140.65	0.0008	0.0005
20	0.10	165.60	-0.63	140.65	0.0006	0.0045
30		165.60	-0.06	136.99		0.0004

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	136.99	0.0000	-0.0000
20	0.16	165.60	-0.06	140.65	0.0010	0.0004
20	0.10	165.60	-0.65	140.65	0.0006	0.0046
30		165.60	-0.06	136.99		0.0004

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-0.00	136.99		0.0000
20	0.15	165.60	-0.06	140.65	0.0009	0.0004
20	0.17	165.60	-0.69	140.65	0.0010	0.0049
30		165.60	-0.06	136.99		0.0004

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	280.80	0.00	136.99	0.0000	-0.0000
20	0.16	280.79	-0.05	140.65	0.0006	0.0003
20		280.79	-0.40	140.65		0.0029
30		280.79	-0.05	136.99		0.0004

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	280.80	0.00	136.99	0.0000	-0.0000
20	0.16	280.79	-0.05	140.65	0.0006	0.0003
20		280.79	-0.28	140.65		0.0020
30		280.79	-0.05	136.99		0.0004

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	43.84	165.60		136.99	0.2647	
20	35.55	165.60		140.65	0.2147	
20	35.50	165.60		140.65	0.2144	
30	43.80	165.60		136.99	0.2645	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	43.84	165.60		136.99	0.2647	
20	35.54	165.60		140.65	0.2146	
20	35.57	165.60		140.65	0.2148	
30	43.80	165.60		136.99	0.2645	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	136.99	0.0000	-0.0000
20	0.16	165.60	-0.06	140.65	0.0010	0.0004
20	0.10	165.60	-0.65	140.65	0.0006	0.0046
30		165.60	-0.06	136.99		0.0004

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-0.00	136.99		0.0000
20	0.15	165.60	-0.06	140.65	0.0009	0.0004
20	0.17	165.60	-0.69	140.65	0.0010	0.0049
30		165.60	-0.06	136.99		0.0004

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	88.89	280.80		136.99	0.3166	
20	71.93	280.79		140.65	0.2562	
20	71.61	280.79		140.65	0.2550	
30	88.85	280.79		136.99	0.3164	



Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	88.89	280.80		136.99	0.3166	
20	71.92	280.79		140.65	0.2561	
20	71.49	280.79		140.65	0.2546	
30	88.84	280.79		136.99	0.3164	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	43.84	165.60		136.99	0.2647	
20	35.53	165.60		140.65	0.2146	
20	35.11	165.60		140.65	0.2120	
30	43.79	165.60		136.99	0.2644	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	43.84	165.60		136.99	0.2647	
20	35.44	165.60		140.65	0.2140	
20	35.15	165.60		140.65	0.2123	
30	43.80	165.60		136.99	0.2645	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	43.84	165.60		136.99	0.2647	
20	35.54	165.60		140.65	0.2146	
20	35.12	165.60		140.65	0.2121	
30	43.79	165.60		136.99	0.2644	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	43.84	165.60		136.99	0.2647	
20	35.53	165.60		140.65	0.2146	
20	35.11	165.60		140.65	0.2120	
30	43.79	165.60		136.99	0.2644	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	136.99	0.0000	-0.0000
20	0.15	165.60	-0.05	140.65	0.0009	0.0004
20		165.60	-0.27	140.65		0.0019
30		165.60	-0.05	136.99		0.0004

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	43.84	138.00		114.16	0.3176	
20	35.54	138.00		117.21	0.2575	
20	35.14	138.00		117.21	0.2546	
30	43.79	138.00		114.16	0.3173	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		138.00	-0.00	114.16		0.0000
20	0.15	138.00	-0.06	117.21	0.0011	0.0005



20	138.00	-0.26	117.21	0.0022
30	138.00	-0.05	114.16	0.0004

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	43.84	165.60		136.99	0.2647	
20	35.54	165.60		140.65	0.2146	
20	35.12	165.60		140.65	0.2121	
30	43.79	165.60		136.99	0.2644	

Absolute Maximum of the all of the Stress Ratio's 0.3176

Governing Element: Btmhead

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Btm_head	43.8	...	...	...	43.8	+ Tens
Btm_head	43.8	...	...	...	...	- Comp
Shell	35.4	...	0.1	...	35.5	+ Tens
Shell	35.4	...	-0.1	...	...	- Comp
Shell	35.4	-0.3	...	...	35.1	+ Tens
Shell	35.4	-0.3	...	...	...	- Comp
Top_head	43.8	...	...	...	43.8	+ Tens
Top_head	43.8	...	...	...	...	- Comp

Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	303.409 mm
Center of Gravity of the Insulation	950.000 mm
Center of Gravity of the Nozzles	565.451 mm
Center of Gravity of the Legs	-442.500 mm
Center of Gravity of the Added Weights (Operating)	950.000 mm
Center of Gravity of the Added Weights (Empty)	950.000 mm
Center of Gravity of Bare Shell New and Cold	950.000 mm
Center of Gravity of Bare Shell Corroded	950.000 mm
Vessel CG in the Operating Condition	702.095 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	804.440 mm
Vessel CG in the Test Condition	868.776 mm

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RESULTS FOR LEGS : Operating Case Description: LEGS

Legs attached to: Shell

Section Properties : Single Angle L3X3X0.2500

USA AISC 1989 Steel Table

Overall Leg Length	1385.000	mm
Effective Leg Length	Leglen 1155.000	mm
Distance Leg Up Side of Vessel	200.000	mm
Number of Legs	Nleg 3	
Cross Sectional Area for L3X3X0.2500	Aleg 929.030	mm ²
Section Inertia (strong axis)	516126.438	mm ⁴
Section Inertia (weak axis)	516126.438	mm ⁴
Section Modulus (strong axis)	9455.335	mm ³
Section Modulus (weak axis)	9455.335	mm ³
Radius of Gyration (strong axis)	23.622	mm
Radius of Gyration (weak axis)	23.622	mm

Leg Orientation - Strong Axis

Overturning Moment at top of Legs	1310107.0	N-mm
Total Weight Load at top of Legs	W 8523.3	N
Total Shear force at top of Legs	1631.6	N
Additional force in Leg due to Bracing	Fadd 0.0	N
Occasional Load Factor	Occfac 1.333	
Effective Leg End Condition Factor	k 1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (1632) (518397 / 1555192)$
 $= 543.88 \text{ N}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W / N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (8524 / 3 + (1309585 / (1 \cdot 331))) / 929$
 $= 7.31 \text{ MPa}$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (8523 / 3) / 929$
 $= 3.06 \text{ MPa}$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	76.200	76.200	6.3500
Distance to geometric centroid:	21.387	21.387	
Arm about YY:	18.212	19.888	
Arm about ZZ:	16.713	18.212	
Leg areas:	483.87	443.55	
Geometric inertia components YY:	0.16211E+06	0.35578E+06	
Geometric inertia components ZZ:	0.36929E+06	0.14860E+06	
Geometric inertias Iy & Iz:	0.51789E+06	0.51789E+06	
Product of inertia:	0.30793E+06		
Mohrs Radius:	0.30793E+06		
Average Inertia:	0.51789E+06		

QFACT =	1.0000	FBZ	=	0.14893
Principal Axis Inertias (Z&W)			=	0.20996E+06 0.82582E+06
Angle to Principal Axis			=	45.000
Distances to extreme fibers CW & CZ			=	53.882 23.636
FOB from Eq 5-5			=	1.0709
Bending allowables Fby & Fbz			=	0.16383 0.14893

Shear Center Coordinates Wo & Zo: 25.368 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E, G, J, R0²: 29500. 11346. 0.30000E-01 2.7256
 AREA, LENGTH, Kw, Kz: 1.4400 45.472 1.0000 1.0000
 H, Few, Fez, Fej: 0.63403 194.01 49.325 86.726
 Fe computed from C4-1: 71.470

Initial (Kl/r)max, & (Kl/r)equiv = 76.830 63.826
 Final (Kl/r)max, & Cc = 76.830 127.18
 Fa based on Eq 4-1 = 0.10877

Leg Bending and Axial Stresses vs. Allowables: (MPa)

	Actual	Allowable
Weak Axis Bending :	50.01	198.53
Strong Axis Bending :	28.98	218.38
Axial Compression :	7.31	144.99

UNITY CHECKS ARE: H1-1 0.000
 H1-2 0.000
 H1-3 0.435

AISC Unity Check : 0.435, should be <= to 1 for Code compliance, [Passed]

Bolting Size Requirement for Leg Baseplates :

Baseplate Material	SA-36
Baseplate Allowable Stress	SBA 150.00 MPa
Baseplate Length	B 154.0000 mm
Baseplate Width	D 154.0000 mm
Baseplate Thickness	BTHK 14.0000 mm
Leg Dimension Along Baseplate Length	d 76.2000 mm
Leg Dimension Along Baseplate Width	b 76.2000 mm
Bolt Circle Diameter	BCD 570.0000 mm
Bolt Material	SA-36
Bolt Allowable Tensile Stress	STBA 150.00 MPa
Bolt Allowable Shear Stress	SBShear 100.00 MPa
Anchor Bolt Nominal Diameter	BOD 24.0000 mm
Number of Anchor Bolts in Tension per Leg	NB 1
Total Number of Anchors Bolt per Leg	NBT 1
Bolt Corrosion Allowance	bca 1.5000 mm
Ultimate 28-day Concrete Strength	FCPRIME 35.000 MPa

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [tb]:
 = Shear Force/(Bolt Area • Number of Bolts)
 = 1632/(226 • 3)
 = 2.4 MPa. Must be less than 100.0 MPa.

Plate thickness required (1/2"; 13mm) per D. Moss

Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Computing the required bolt area based on a standard method not including the baseplate's interaction with the foundation.

Calculation of Load per Bolt, Wind Empty [W,wind]:
 = ((4 • Moment/BCD) - Weight)/Num Bolts
 = ((4 • 1142606/570) - 7348)/3
 = 222.485 N

Calculation of Load per Bolt, Seismic Operating [W,seismic]:
 = ((4 • Moment/BCD) - Weight)/Num Bolts
 = ((4 • 1310107/570) - 8524)/3



= 222.059 N

There was a vertical load reducing the weight on the support.

Calculation of Maximum Load per Bolt [Wmax]:

= max(W_{wind}, W_{seismic})

= max(222, 222)

= 222.485 N

Area Available in a Single Bolt, corroded [Ab]:

= 225.781 mm²

Required Area of a single Bolt, Simplified Method:

= Wmax / Bolt Allowable Stress

= 222/14035391

= 1.483 mm²

Note: The Wind case empty condition governs.

[Review notes about nozzle loadings and supports in the Warnings report.](#)

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RESULTS FOR LEGS : HydroTest Case Description: LEGS

Legs attached to: Shell

Section Properties : Single Angle L3X3X0.2500

USA AISC 1989 Steel Table

Overall Leg Length	1385.000	mm
Effective Leg Length	Leglen 1155.000	mm
Distance Leg Up Side of Vessel	200.000	mm
Number of Legs	Nleg 3	
Cross Sectional Area for L3X3X0.2500	Aleg 929.030	mm ²
Section Inertia (strong axis)	516126.438	mm ⁴
Section Inertia (weak axis)	516126.438	mm ⁴
Section Modulus (strong axis)	9455.335	mm ³
Section Modulus (weak axis)	9455.335	mm ³
Radius of Gyration (strong axis)	23.622	mm
Radius of Gyration (weak axis)	23.622	mm

Leg Orientation - Strong Axis

Overturning Moment at top of Legs	377060.0	N-mm
Total Weight Load at top of Legs	W 12055.1	N
Total Shear force at top of Legs	538.4	N
Additional force in Leg due to Bracing	Fadd 0.0	N
Occasional Load Factor	Occfac 1.333	
Effective Leg End Condition Factor	k 1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (538) (518397 / 1555192)$
 $= 179.48 \text{ N}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W / N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (12056 / 3 + (376910 / (1 \cdot 331))) / 929$
 $= 5.55 \text{ MPa}$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (12055 / 3) / 929$
 $= 4.33 \text{ MPa}$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	76.200	76.200	6.3500
Distance to geometric centroid:	21.387	21.387	
Arm about YY:	18.212	19.888	
Arm about ZZ:	16.713	18.212	
Leg areas:	483.87	443.55	
Geometric inertia components YY:	0.16211E+06	0.35578E+06	
Geometric inertia components ZZ:	0.36929E+06	0.14860E+06	
Geometric inertias Iy & Iz:	0.51789E+06	0.51789E+06	
Product of inertia:	0.30793E+06		
Mohrs Radius:	0.30793E+06		
Average Inertia:	0.51789E+06		

QFACT =	1.0000	FBZ	=	0.14893
Principal Axis Inertias (Z&W)			=	0.20996E+06 0.82582E+06
Angle to Principal Axis			=	45.000
Distances to extreme fibers CW & CZ			=	53.882 23.636
FOB from Eq 5-5			=	1.0709
Bending allowables Fby & Fbz			=	0.16383 0.14893

Shear Center Coordinates Wo & Zo: 25.368 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E,	G,	J,	R0 ² :	29500.	11346.	0.30000E-01	2.7256
AREA,	LENGTH,	Kw,	Kz:	1.4400	45.472	1.0000	1.0000
H,	Few,	Fez,	Fej:	0.63403	194.01	49.325	86.726
Fe computed from C4-1:				71.470			
Initial (Kl/r)max, & (Kl/r)equiv				=	76.830	63.826	
Final (Kl/r)max, & Cc				=	76.830	127.18	
Fa based on Eq 4-1				=	0.10877		

Leg Bending and Axial Stresses vs. Allowables: (MPa)

	Actual	Allowable
Weak Axis Bending :	16.50	198.53
Strong Axis Bending :	9.56	218.38
Axial Compression :	5.55	144.99

UNITY CHECKS ARE: H1-1 0.000
 H1-2 0.000
 H1-3 0.165

AISC Unity Check : 0.165, should be <= to 1 for Code compliance, [Passed]

Bolting Size Requirement for Leg Baseplates :

Baseplate Material		SA-36
Baseplate Allowable Stress	SBA	150.00 MPa
Baseplate Length	B	154.0000 mm
Baseplate Width	D	154.0000 mm
Baseplate Thickness	BTHK	14.0000 mm
Leg Dimension Along Baseplate Length	d	76.2000 mm
Leg Dimension Along Baseplate Width	b	76.2000 mm
Bolt Circle Diameter	BCD	570.0000 mm
Bolt Material		SA-36
Bolt Allowable Tensile Stress	STBA	150.00 MPa
Bolt Allowable Shear Stress	SBShear	100.00 MPa
Anchor Bolt Nominal Diameter	BOD	24.0000 mm
Number of Anchor Bolts in Tension per Leg	NB	1
Total Number of Anchors Bolt per Leg	NBT	1
Bolt Corrosion Allowance	bca	1.5000 mm
Ultimate 28-day Concrete Strength	FCPRIME	35.000 MPa

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [tb]:
 = Shear Force/(Bolt Area • Number of Bolts)
 = 1632/(226 • 3)
 = 2.4 MPa. Must be less than 100.0 MPa.

Plate thickness required (1/2"; 13mm) per D. Moss

Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: B22"

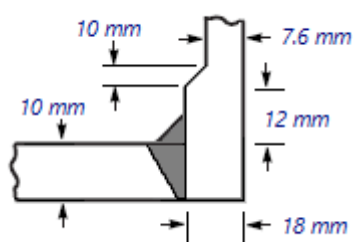
From: 10

Pressure for Reinforcement Calculations	P	3.9053	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	600.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	12.0000	mm
Height of Beveled Transition	L'	10.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		18.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: B22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.91 \cdot 603 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 3.91)$
 $= 7.6643 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.91 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 3.91)$
 $= 0.8110 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 10.4 / 0.5773$
 $= 17.9347 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	96.0684 mm
Parallel to Vessel Wall, opening length	d	48.0342 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	21.2500 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: B22".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.3110 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 10.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 10.0000 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 4.9200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(10, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.31, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
 (see 319.4.4 and 302.3.5):**

Sustained	:	44.4,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	250.6 MPa	Passed
Occasional	:	6.8,	Allowable	:	156.9 MPa	Passed
Shear	:	30.2,	Allowable	:	82.6 MPa	Passed

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of
 the expansion stress allowable.*



Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: B22"

Intermediate Calc. for nozzle/shell Welds Tmin 10.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.4999 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 261.4999 mm

Input Echo, WRC107/537 Item 1, Description: B22" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm
Vessel Diameter	Dv	1080.000	mm
Vessel Thickness	Tv	10.000	mm
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm
Nozzle Diameter	Dn	45.034	mm
Nozzle Thickness	Tn	18.000	mm
Nozzle Material		SA-350 LF1	
Nozzle UNS Number		K03009	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Design Internal Pressure	Dp	3.905	MPa
Include Pressure Thrust		No	

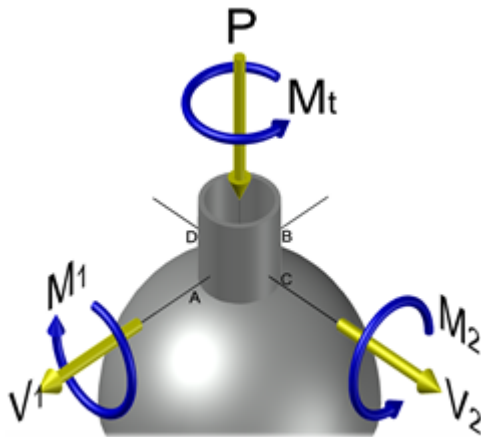
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-3199.9	N
Longitudinal Shear (SUS)	(V1) V1	2828.3	N
Circumferential Shear (SUS)	(Vc) V2	2828.3	N
Circumferential Moment (SUS)	(Mc) M1	371220.7	N-mm
Longitudinal Moment (SUS)	(Ml) M2	371220.7	N-mm
Torsional Moment (SUS)	Mt	480169.5	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 81.034 + 2 \cdot 1.65 \cdot \sqrt{(545.75 (10.0 - 1.5))}$$

$$= 305.795 \text{ mm}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load		P	-3199.9	N
Circumferential Shear	(VC)	V2	2828.3	N
Longitudinal Shear	(VL)	V1	2828.3	N
Circumferential Moment	(MC)	M1	371220.7	N-mm
Longitudinal Moment	(ML)	M2	371220.7	N-mm
Torsional Moment		MT	480169.5	N-mm

Dimensionless Param: U = 0.59 TAU = 5.00 (1.96) RHO = 0.52

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.06520	(A,B,C,D)
$M(x) / P$	SP 1	0.07743	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.08804	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.18273	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.08804	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.18273	(A,B,C,D)
$N(y) * T / P$	SP 1	0.10692	(A,B,C,D)
$M(y) / P$	SP 1	0.02385	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.09587	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.06495	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.09587	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.06495	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Rad. Memb. P	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9
Rad. Bend. P	20.6	-20.6	20.6	-20.6	20.6	-20.6	20.6	-20.6
Rad. Memb. MC	0.0	0.0	0.0	0.0	-6.6	-6.6	6.6	6.6
Rad. Bend. MC	0.0	0.0	0.0	0.0	-82.7	82.7	82.7	-82.7
Rad. Memb. ML	-6.6	-6.6	6.6	6.6	0.0	0.0	0.0	0.0
Rad. Bend. ML	-82.7	82.7	82.7	-82.7	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-65.9	58.4	112.8	-93.7	-65.9	58.4	112.8	-93.7
Tang. Memb. P	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7
Tang. Bend. P	6.3	-6.3	6.3	-6.3	6.3	-6.3	6.3	-6.3
Tang. Memb. MC	0.0	0.0	0.0	0.0	-7.2	-7.2	7.2	7.2
Tang. Bend. MC	0.0	0.0	0.0	0.0	-29.4	29.4	29.4	-29.4
Tang. Memb. ML	-7.2	-7.2	7.2	7.2	0.0	0.0	0.0	0.0
Tang. Bend. ML	-29.4	29.4	29.4	-29.4	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-25.5	20.6	47.7	-23.8	-25.5	20.6	47.7	-23.8
Shear VC	2.6	2.6	-2.6	-2.6	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-2.6	-2.6	2.6	2.6
Shear MT	5.5	5.5	5.5	5.5	5.5	5.5	5.5	5.5
Tot. Shear	8.1	8.1	2.9	2.9	2.9	2.9	8.1	8.1
Str. Int.	67.4	60.0	112.9	93.9	66.1	58.6	113.8	94.7

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		124.4	124.4	124.4	124.4	124.4	124.4	124.4	124.4
Rad. Pl (SUS)		-3.8	-3.8	9.5	9.5	-3.8	-3.8	9.5	9.5
Rad. Q (SUS)		-62.1	62.1	103.3	-103.3	-62.1	62.1	103.3	-103.3
Long. Pm (SUS)		124.4	124.4	124.4	124.4	124.4	124.4	124.4	124.4
Long. Pl (SUS)		-2.5	-2.5	12.0	12.0	-2.5	-2.5	12.0	12.0
Long. Q (SUS)		-23.1	23.1	35.7	-35.7	-23.1	23.1	35.7	-35.7
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		2.6	2.6	-2.6	-2.6	-2.6	-2.6	2.6	2.6
Shear Q (SUS)		5.5	5.5	5.5	5.5	5.5	5.5	5.5	5.5
Pm (SUS)		124.4	124.4	124.4	124.4	124.4	124.4	124.4	124.4
Pm+Pl (SUS)		124.0	124.0	138.0	138.0	124.0	124.0	138.0	138.0
Pm+Pl+Q (Total)		100.4	184.4	237.3	100.8	99.1	183.0	238.2	101.6

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	124.41	138.00	Passed
Pm+Pl (SUS)	138.04	207.00	Passed
Pm+Pl+Q (TOTAL)	238.19	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: A12"

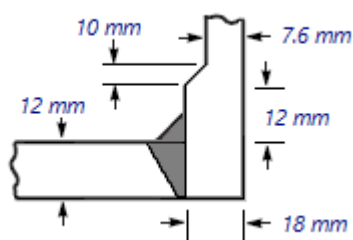
From: 20

Pressure for Reinforcement Calculations	P	3.8995	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	600.00	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1050.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	12.0000	mm
Height of Beveled Transition	L'	10.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		18.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: A12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.9 \cdot 302 / (138 \cdot 1 - 0.6 \cdot 3.9)$
 $= 8.6665 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.9 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 3.9)$
 $= 0.8097 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 10.4 / 0.5773$
 $= 17.9347 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	96.0684	mm
Parallel to Vessel Wall, opening length	d	48.0342	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: A12".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.3097	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 10.1665	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 10.1665	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(10.2, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.31, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	44.4,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	250.6 MPa	Passed
Occasional	:	6.8,	Allowable	:	156.9 MPa	Passed
Shear	:	30.2,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.



Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: A12"

Intermediate Calc. for nozzle/shell Welds Tmin 10.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 2.7487 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.7487 mm

Input Echo, WRC297 Item 1, Description: A12" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	600.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	45.034	mm
Nozzle Thickness	18.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	3.90	MPa
Radial Load	P	-3199.86	N
Circumferential Shear	Vc	2399.89	N
Longitudinal Shear	Vl	3199.86	N
Circumferential Moment	Mc	320112.56	N-mm
Longitudinal Moment	Ml	416146.78	N-mm
Torsional Moment	Mt	480169.53	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.502) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 81.034 + 2 \cdot 1.65 \cdot \sqrt{306.75(12.0 - 1.5)}$$

$$= 268.318 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	613.500 mm
Nozzle Outside Diameter	(d) =	81.034 mm
Vessel Thickness used	(T) =	10.500 mm
Nozzle Thickness used	(t) =	16.500 mm
T / t	=	0.636
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 1.010$$

Nr/P	=	0.110
Mr/P	=	0.213
M0/P	=	0.065
N0/P	=	0.130
MrD/Mc	=	0.299
NrDL/Mc	=	0.106
M0d/Mc	=	0.099
N0DL/Mc	=	0.018
MrD/Ml	=	0.196
NrDL/Ml	=	0.094
M0D/Ml	=	0.066
N0DL/Ml	=	0.119

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	110	113	110	113
Outplane Stress Summary	101	119	149	93

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-54	54	54	-54
Inplane Pressure Stress	55	55	55	55
Inplane Stress Summary	36	71	154	-29

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary 6 6 2 2

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 101 120 155 122

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-21	21	21	-21
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	55	55	55	55

Outplane Stress Summary 48 68 92 26

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-64	64	64	-64
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	110	113	110	113

Inplane Stress Summary 82 140 218 19

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary 2 2 6 6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress 82 140 218 30

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses in the
 hoop direction)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	9	5	9



Hoop Stress Summary 3 7 14 19

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9	-9	9	9
Axial Bending (ML)	-16	16	16	-16
Axial Pressure Stress	2	2	2	2

Axial Stress Summary -11 0 40 -15

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress)

	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary 3 3 1 1

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress 16 11 40 34

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses in the
 hoop direction)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	9	5	9

Hoop Stress Summary 8 12 9 14

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	-6	-6	6	6
Axial Bending (Mc)	-21	21	21	-21
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary -13 6 42 -22

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1

Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	4	4

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	22	13	43	37

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	110	113	110	113	110	113	110	113
Circ. Pl (SUS)	-2	-2	8	8	0	0	6	6
Circ. Q (SUS)	-7	7	29	-29	-27	27	101	-101
Long. Pm (SUS)	55	55	55	55	55	55	55	55
Long. Pl (SUS)	-1	-1	7	7	3	3	3	3
Long. Q (SUS)	-17	17	91	-91	-10	10	32	-32
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-2	-2	2	2
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	110.0	113.0	110.0	113.0	110.0	113.0	110.0	113.0
Pm+Pl (SUS)	108.0	111.0	118.0	121.0	110.1	113.1	116.1	119.1
Pm+Pl+Q (Total)	101.4	118.5	154.2	121.1	83.1	140.1	217.3	29.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	113.00	138.00	Passed
Pm+Pl (SUS)	121.02	207.00	Passed
Pm+Pl+Q (TOTAL)	217.28	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	9	5	9	5	9	5	9
Circ. Pl (SUS)	-2	-2	8	8	3	3	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-9	-9	9	9	-6	-6	6	6
Long. Q (SUS)	-5	5	27	-27	-10	10	32	-32
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	5.0	9.0	5.0	9.0	5.0	9.0	5.0	9.0
Pm+Pl (SUS)	10.2	14.1	13.4	17.2	12.2	16.1	9.0	12.2



Pm+Pl+Q (Total)	16.2	10.8	38.0	33.1	22.1	12.2	40.3	36.5
-----------------	------	------	------	------	------	------	------	------

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	118.00	Passed
Pm+Pl (SUS)	17.16	177.00	Passed
Pm+Pl+Q (TOTAL)	40.28	354.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: H16"

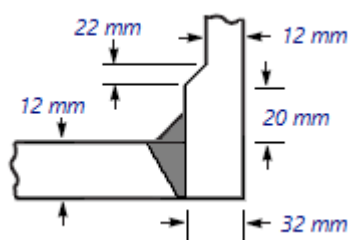
From: 20

Pressure for Reinforcement Calculations	P	3.9029	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	600.00	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		200.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	20.0000	mm
Height of Beveled Transition	L'	22.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		32.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: H16"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 6.000 in.
 Actual Thickness Used in Calculation 0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.9 \cdot 302 / (138 \cdot 1 - 0.6 \cdot 3.9)$
 $= 8.6744$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.9 \cdot 77.7 / (118 \cdot 1 - 0.6 \cdot 3.9)$
 $= 2.6220$ mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 20 / 0.5773$
 $= 34.6410$ mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

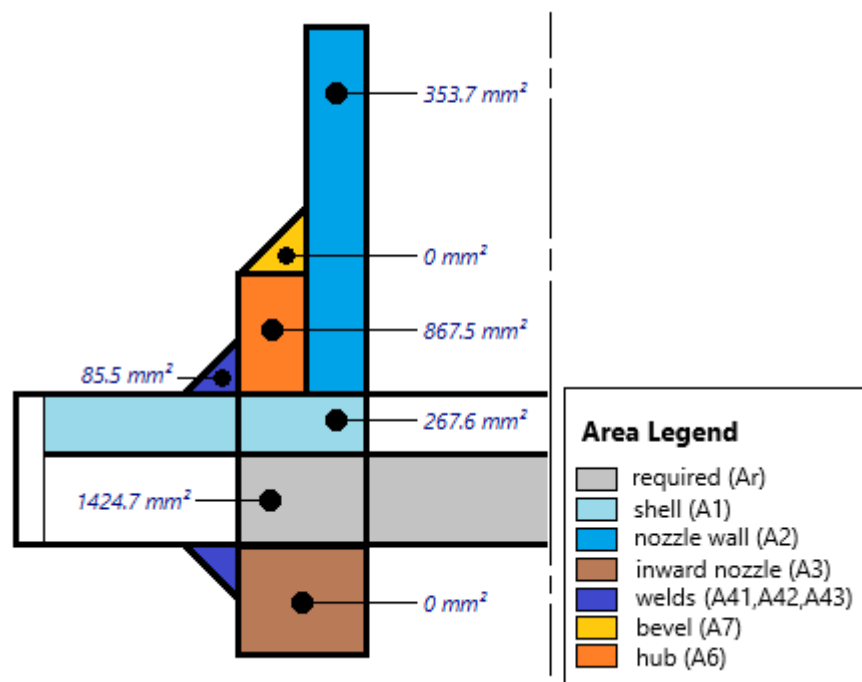
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	310.8000	mm
Parallel to Vessel Wall, opening length	d	155.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1424.682	NA	NA
Area in Shell	A1	267.559	NA	NA
Area in Nozzle Wall	A2	353.659	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.509	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	867.477	NA	NA
TOTAL AREA AVAILABLE	Atot	1574.204	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (155 \cdot 8.67 \cdot 1 + 2 \cdot 30.5 \cdot 8.67 \cdot 1 \cdot (1 - 0.86))$
 $= 1424.682 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 155 (1 \cdot 10.5 - 1 \cdot 8.67) - 2 \cdot 30.5$
 $(1 \cdot 10.5 - 1 \cdot 8.67) \cdot (1 - 0.86)$
 $= 267.559 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 26.2) (10.5 - 2.62) 0.86$
 $= 353.659 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$
 $= 85.509 \text{ mm}^2$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(26.2, 200, 20)) \cdot (32 - 12) \cdot 0.86$$

$$= 867.477 \text{ mm}^2 \text{ Includes Bevel Area of } 183.407 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.1220 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 3.0000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 10.1744 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 10.1744 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 8.6600 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[8.66, \max(10.2, 3)]$$

$$= 8.6600 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.12, 8.66)$$

$$= 8.6600 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-34 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $3.90/4.26 = 0.916$

Weld Size Calculations, Description: H16"

Intermediate Calc. for nozzle/shell Welds	T_{min}	10.5000 mm
---	-----------	------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - tr)) S_v)$$

$$= \max(0, (1425 - 268 + 2 \cdot 30.5 \cdot 0.86 \cdot (1 \cdot 10.5 - 8.67)) 138)$$

$$= 172806.00 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - C_{an} / .707)^2 \cdot f_r2) \cdot S_v$$

$$= (1221 + 85.5 - 0 \cdot 0.86) \cdot 138$$

$$= 180298.47 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$$

$$= (1221 + 0 + 85.5 + (548)) \cdot 138$$

$$= 255871.05 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (1221 + 0 + 85.5 + 0 + (548)) \cdot 138$$

$$= 255871.05 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 216 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 196525. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - C_{an}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 92.9) \cdot (32 - 1.5) \cdot 0.7 \cdot 118$$

$$= 735601. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 216 \cdot (8 - 0) \cdot 0.74 \cdot 138$$

$$= 277673. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (196525 + 735601) = 932126 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (196525 + 0 + 277673 + 0) = 474198 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (196525 + 277673 + 0) = 474198 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 932126 N	, must exceed W = 172806 N	or W1 = 180298 N
Path 2-2 = 474197 N	, must exceed W = 172806 N	or W2 = 255871 N
Path 3-3 = 474197 N	, must exceed W = 172806 N	or W3 = 255871 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4.11 MPa

The Drop for this Nozzle is : 20.1916 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 232.1916 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 155.40 mm

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Input, Nozzle Desc: L12"

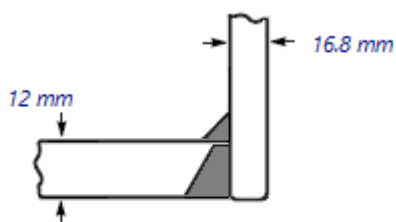
From: 20

Pressure for Reinforcement Calculations	P	3.8995	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	600.00	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		800.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.9 \cdot 302 / (138 \cdot 1 - 0.6 \cdot 3.9)$
 $= 8.6665 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([3.9 / (118 \cdot 1)] - 1))$
 $= 0.9038 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.6000	mm
Parallel to Vessel Wall, opening length	d	53.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L12".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.4038	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 10.1665	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 10.1665	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(10.2, 3)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.4, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Weld Size Calculations, Description: L12"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.5000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 2.9778 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.3528 mm



This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 53.80 mm

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Input, Nozzle Desc: L32"

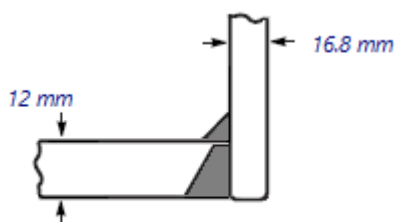
From: 20

Pressure for Reinforcement Calculations	P	3.8995	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	600.00	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		800.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.9 \cdot 302 / (138 \cdot 1 - 0.6 \cdot 3.9)$
 $= 8.6665 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([3.9 / (118 \cdot 1)] - 1))$
 $= 0.9038 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.6000	mm
Parallel to Vessel Wall, opening length	d	53.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L32".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.4038	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 10.1665	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 10.1665	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(10.2, 3)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.4, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Weld Size Calculations, Description: L32"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.5000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 2.9778 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.3528 mm





Input, Nozzle Desc: L22"

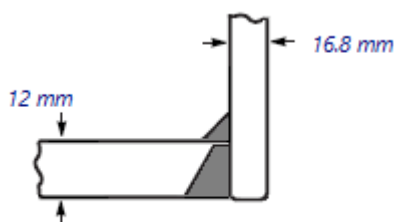
From: 20

Pressure for Reinforcement Calculations	P	3.9016	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	600.00	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		400.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.9 \cdot 302 / (138 \cdot 1 - 0.6 \cdot 3.9)$
 $= 8.6714 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([3.9 / (118 \cdot 1)] - 1))$
 $= 0.9043 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.6000	mm
Parallel to Vessel Wall, opening length	d	53.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L22".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.4043	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 10.1714	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 10.1714	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(10.2, 3)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.4, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Weld Size Calculations, Description: L22"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.5000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 2.9778 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.3528 mm



Input, Nozzle Desc: L4"2"

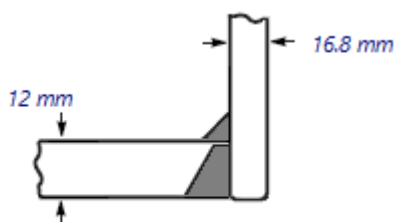
From: 20

Pressure for Reinforcement Calculations	P	3.9016	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	600.00	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		400.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L4"2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.9 \cdot 302 / (138 \cdot 1 - 0.6 \cdot 3.9)$
 $= 8.6714 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([3.9 / (118 \cdot 1)] - 1))$
 $= 0.9043 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.6000	mm
Parallel to Vessel Wall, opening length	d	53.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L4"2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.4043	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 10.1714	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 10.1714	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(10.2, 3)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.4, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Weld Size Calculations, Description: L4"2"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.5000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 2.9778 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.3528 mm





Input, Nozzle Desc: B13"

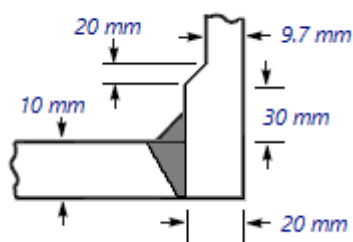
From: 30

Pressure for Reinforcement Calculations	P	3.8995	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	600.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	30.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		20.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: B13"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.734 in.
 Actual Thickness Used in Calculation 0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.9 \cdot 603 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 3.9)$
 $= 7.6530 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.9 \cdot 36.2 / (118 \cdot 1 - 0.6 \cdot 3.9)$
 $= 1.2210 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 10.3 / 0.5773$
 $= 17.7803 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	144.8618	mm
Parallel to Vessel Wall, opening length	d	72.4309	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	21.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: B13".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.7210	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 9.9875	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 9.9875	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(9.99, 3)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.72, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	35.7,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	259.3 MPa	Passed
Occasional	:	7.7,	Allowable	:	156.9 MPa	Passed
Shear	:	19.5,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-68 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $3.90/8.52 = 0.458$

Weld Size Calculations, Description: B13"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.5000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.91 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.7384 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 212.7384 mm

Input Echo, WRC107/537 Item 1, Description: B13" :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	1.5000 mm
Vessel Diameter	Dv	1080.000 mm
Vessel Thickness	Tv	10.000 mm
Design Temperature	T1	85.0 C
Vessel Material		SA-516 70
Vessel UNS Number		K02700
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	ID
Corrosion Allowance for Nozzle	Can	1.5000 mm

Nozzle Diameter	Dn	69.431	mm
Nozzle Thickness	Tn	20.000	mm
Nozzle Material		SA-350 LF1	
Nozzle UNS Number		K03009	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

Design Internal Pressure	Dp	3.899	MPa
Include Pressure Thrust		No	

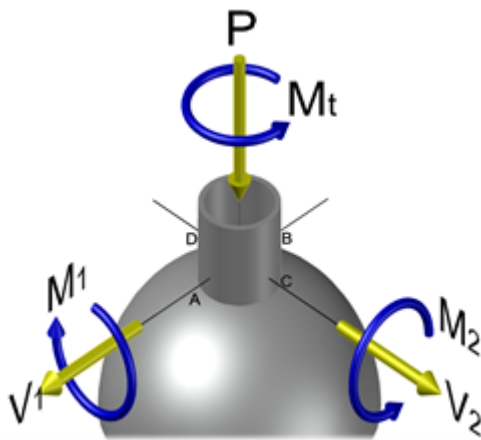
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	-4200.0	N
Longitudinal Shear	(SUS)	(V1) V1	4200.0	N
Circumferential Shear	(SUS)	(Vc) V2	3150.0	N
Circumferential Moment	(SUS)	(Mc) M1	819000.4	N-mm
Longitudinal Moment	(SUS)	(Ml) M2	819000.4	N-mm
Torsional Moment	(SUS)	Mt	944999.4	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))} \\
 &= 109.431 + 2 \cdot 1.65 \cdot \sqrt{(545.75 (10.0 - 1.5))} \\
 &= 334.192 \text{ mm}
 \end{aligned}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-4200.0	N
Circumferential Shear	(VC) V2	3150.0	N
Longitudinal Shear	(VL) V1	4200.0	N
Circumferential Moment	(MC) M1	819000.4	N-mm
Longitudinal Moment	(ML) M2	819000.4	N-mm
Torsional Moment	MT	944999.4	N-mm

Dimensionless Param: U = 0.80 TAU = 5.00 (2.46) RHO = 0.46

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
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N(y) * T / P	SP 1	0.08207	(A,B,C,D)
M(y) / P	SP 1	0.01657	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 1	0.08123	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 1	0.03850	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 1	0.08123	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 1	0.03850	(A,B,C,D)

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9
Rad.	Bend. P	19.5	-19.5	19.5	-19.5	19.5	-19.5	19.5	-19.5
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-10.6	-10.6	10.6	10.6
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-114.6	114.6	114.6	-114.6
Rad.	Memb. ML	-10.6	-10.6	10.6	10.6	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-114.6	114.6	114.6	-114.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-102.8	87.4	147.7	-120.5	-102.8	87.4	147.7	-120.5
Tang. Memb. P		4.8	4.8	4.8	4.8	4.8	4.8	4.8	4.8
Tang. Bend. P		5.8	-5.8	5.8	-5.8	5.8	-5.8	5.8	-5.8
Tang. Memb. MC		0.0	0.0	0.0	0.0	-13.5	-13.5	13.5	13.5
Tang. Bend. MC		0.0	0.0	0.0	0.0	-38.4	38.4	38.4	-38.4
Tang. Memb. ML		-13.5	-13.5	13.5	13.5	0.0	0.0	0.0	0.0
Tang. Bend. ML		-38.4	38.4	38.4	-38.4	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-41.4	23.9	62.5	-25.9	-41.4	23.9	62.5	-25.9
Shear VC		2.2	2.2	-2.2	-2.2	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.9	-2.9	2.9	2.9
Shear MT		5.9	5.9	5.9	5.9	5.9	5.9	5.9	5.9
Tot. Shear		8.1	8.1	3.8	3.8	3.0	3.0	8.8	8.8
Str. Int.		103.9	88.4	147.8	120.7	103.0	87.5	148.6	121.4

Vessel Stress Summation at Attachment Junction (MPa)

[illegible]



Pm+Pl (SUS)	118.2	118.2	143.4	143.4	118.9	118.9	143.9	143.9
Pm+Pl+Q (Total)	83.9	212.6	272.1	98.5	83.0	211.8	272.8	99.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	124.23	138.00	Passed
Pm+Pl (SUS)	143.88	207.00	Passed
Pm+Pl+Q (TOTAL)	272.81	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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